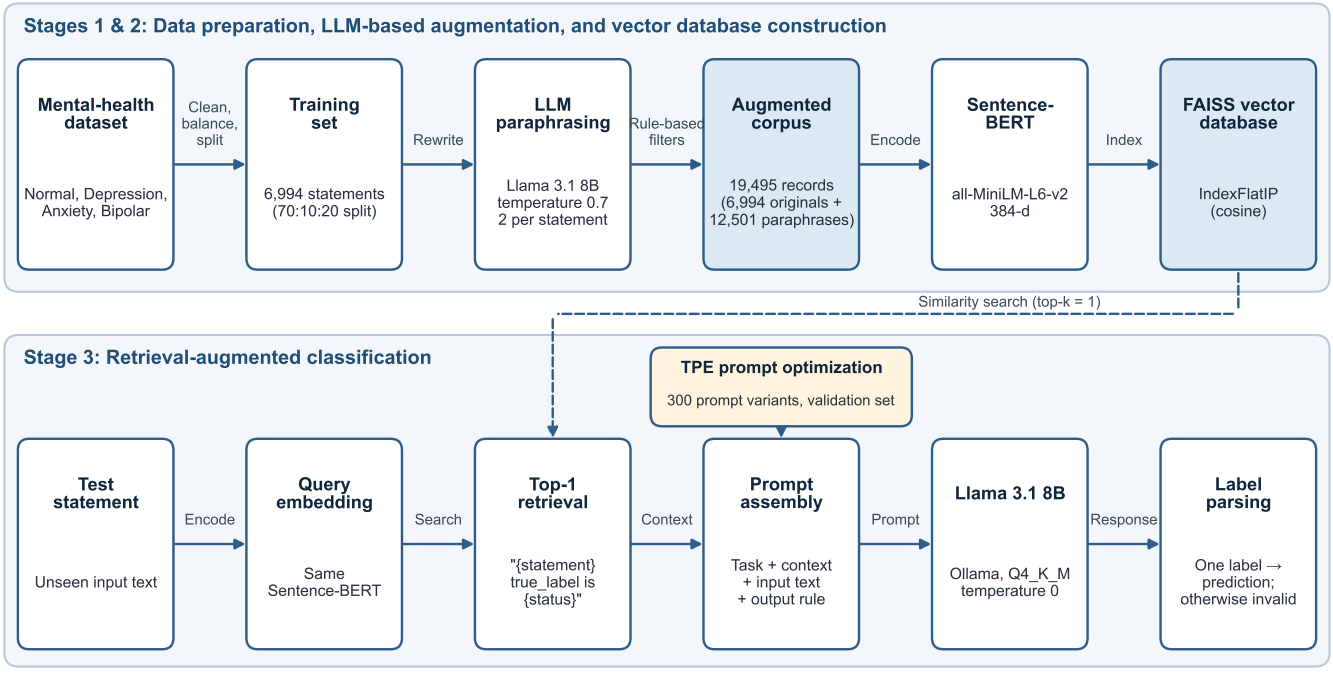
Stages 1 & 2: Data preparation, LLM-based augmentation, and vector database construction
Mental-health dataset
Normal, Depression, Anxiety, Bipolar
Clean, balance, split
Training set
6,994 statements (70:10:20 split)
Rewrite
LLM paraphrasing
Llama 3.1 8B temperature 0.7 2 per statement
Rule-based filters
Augmented corpus
19,495 records (6,994 originals + 12,501 paraphrases)
Encode
Sentence-BERT
all-MiniLM-L6-v2 384-d
Index
FAISS vector database
IndexFlatIP (cosine)
Similarity search (top-k = 1)
Stage 3: Retrieval-augmented classification
TPE prompt optimization
300 prompt variants, validation set
Test statement
Unseen input text
Encode
Query embedding
Same Sentence-BERT
Search
Top-1 retrieval
"{statement} true_label is {status}"
Context
Prompt assembly
Task + context + input text + output rule
Prompt
Llama 3.1 8B
Ollama, Q4_K_M temperature 0
Response
Label parsing
One label → prediction; otherwise invalid